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Stable isotope data from loess malacofauna: Evidence for climate changes in the Pannonian Basin during the Late Pleistocene

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ABSTRACT

Loess is terrestrial clastic sediment, composed dominantly of silt-sized particles formed by the accumulation of wind-blown dust. It is usually inter-bedded with palaeosol horizons forming loess-palaeosol successions (LPS). Due to its characteristics loess represents a high quality record of climate changes in the Late Pleistocene. The thickest and best-studied loess sections in the Republic of Croatia are located in the eastern Croatian province of Baranja, near the Danube River. This area is a southern part of the Pannonian Basin. Zmajevac LPS was selected in this study. Grain-size distribution indicates that the loess from Zmajevac LPS in Baranja is typical loess, comparable with other loess profiles in the Pannonian Basin.

The faunal assemblages in the loess samples prove the dominance of transitional palaeoecological assemblage types, whereas uniformly defined types are rare. *Helicopsis striata* assemblage is dominant, while *Columella columella* and *Chondrula tridens* assemblages are present in smaller proportion. For additional information about palaeoclimate and palaeoecology in Pannonian Basin, stable oxygen and carbon isotope ratios were measured in mollusc shells. Stable oxygen values range from −5.76‰ to −2.45‰. Stable carbon isotope values range from −8.83‰ to −6.84‰. Stable oxygen values show significant palaeotemperature changes during the Late Pleistocene, with the average growing season (AGS) temperature range of 13.24 °C or 9.457 °C, depending on which formula is applied. The overall climate was much cooler than present. Stable carbon isotope values point to dominance of C3 vegetation type during the Late Pleistocene in southern part of the Pannonian Basin.

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1. Introduction

Loess covers up to 10% of the world's surface area and is usually inter-bedded with palaeosol horizons forming loess-palaeosol sequences, or LPS (Pecsi, 1968). Such successions provide very detailed insight into Pleistocene climatic fluctuations (Xiao et al., 1995; Vanderberghe and Nugteren, 2001; Marković et al., 2007, 2008; Novotny et al., 2011). Due to this characteristic, LPS's are excellent records of palaeoclimate and environmental changes in the Pannonian Basin of the last 1 Ma (Marković et al., 2011). The region of Baranja in north-eastern Croatia (Fig. 1), located in the southern edge of Pannonian Basin is almost completely covered

with Pleistocene and Holocene sediments. In particular, some of the thickest loess successions of Croatia are exposed along the Danube River in Baranja. The total thickness of these loess and loess-like deposits probably exceeds 50 m (Banak, 2012).

LPS of Baranja and northeastern Croatia have a long history of investigation (Galović et al., 2009; Wacha and Frechen, 2011 and references therein). Periglacial phenomena in eastern Croatia were analysed almost five decades ago (Malez, 1965). LPS and their mollusc fauna were investigated, providing an overview on warm periods in the Late Pleistocene of northeastern Croatia (Rukavina, 1983). Others focused on the molluscan fauna from LPS at the Vukovar and Đakovo loess plateaus, just south of the Baranja region (Poje, 1982, 1985, 1986).

The aim of this study is a reconstruction of climate changes during the Late Pleistocene based on the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values measured in land-snail shells from loess samples of the

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Fig. 1. Position of Baranja region at the southern edge of the Pannonian Basin and investigated Zmajevac LPS marked with a black circle. This position map is made by using a global multi-resolution topography map (Ryan et al., 2009, <http://www.geomapp.org>)

Zmajevac LPS. Emphasis will be on $\delta^{18}\text{O}$ values which will be used for determination of paleotemperature changes. The values of stable isotopes of oxygen ($\delta^{18}\text{O}$) and carbon ($\delta^{13}\text{C}$) can be used as a palaeothermometer, as characteristic mollusc assemblages (Ložek, 1964; Krolopp and Sümegei, 1995), or in a broader sense, as a tool to reconstruct the climatic conditions of the Late Pleistocene (Yapp, 1979; Lecolle, 1985; Goodfriend and Magaritz, 1987; Goodfriend, 1992). This stable isotope data will be integrated with grain size analyses and mollusc assemblages to provide a better control on palaeoclimate interpretations. Previous research in the Pannonian Basin did not include stable isotope analysis of mollusc shells for palaeoclimatic reconstructions in this time frame. The only paper which included this analysis as a part of a comprehensive loess study in this region is limited to the narrow time span represented by the Last glacial maximum (LGM), which is a part of MIS2 and only to $\delta^{18}\text{O}$ values (Kehrwald et al., 2010). Most researchers used only mollusc assemblages as a malacothermometer tool, establishing mean annual temperature (MAT) values and/or average summer month temperatures represented by mean July temperatures – MJT (Krolopp and Sümegei, 1995; Sümegei and Krolopp, 2002; Novotny et al., 2011). Most recently, X-ray fluorescence (XRF) and magnetic susceptibility (MS) based palaeoclimatic results have been established in the Pannonian Basin and they determined the palaeotemperatures in the 6.7–8.9 °C range (Schatz et al., 2014). However, other researchers in Southern (Colonese et al., 2007) and Western Europe (Lecolle, 1985; Yanes et al., 2008, 2009), Eastern Mediterranean (Goodfriend and Magaritz, 1987; Goodfriend, 1992, 1999) and also in North America (Yapp, 1979; Balakrishnan et al., 2005) used stable isotope analysis widely in the last three decades. All of these results will serve as a good marker for the present study. The most detailed explanation of the flux-balance model, or physical and chemical processes during which isotope fractionation occurs, and subsequent incorporation of that isotope

ratios in mollusc shells is given by Balakrishnan and Yapp (2004). The authors gave insight into possible interpretational mistakes and constraints of this method when used in palaeoclimatic research.

2. Geological setting

The geology of Baranja (Fig. 2) is characterised by dominantly Pleistocene and Holocene sediments, with some outcrops of Miocene igneous rocks, limestones and marls (CGS, 2009). Holocene sediments are alluvial gravel and gravel, sand and silt deposited in a marsh environment. Pleistocene sediments are dominantly loess, loess-like sediment and palaeo – Danube flood plain sediments. Loess-like sediment represents reworked loess or/and aeolian silt deposited in an aquatic environment (Pecsi, 1990). The only topographic high in the Baranja region is the Bansko Brdo hill (244 m asl), a tectonic fault block, situated in the northern part of the region, stretching 20 km in an NE–SW direction and reaching banks of the Danube river. The combination of active neotectonics (Hećimović, 1991) and Danube river erosion exposed big outcrops of Middle and Upper Pleistocene sediments (loess-palaeosol sequences-LPS) at the surface (Fig. 2).

The oldest exposed rocks at Bansko Brdo hill belong to the Miocene volcano–sedimentary complex, and include basalt-andesite and pyroclastic rocks comprising volcanic and tuffaceous breccias and conglomerates (Fig. 2). K–Ar radiometric, whole-rock measurements indicate an early Middle Miocene age (13.8 ± 0.4 and 14.5 ± 0.4 Ma, Pamić and Pécskay, 1996). The basalt-andesite intercalates Middle Miocene (Badenian) marine calcareous sand and marl, indicating a synsedimentary character of the volcanic activity (Pamić and Pikija, 1987; Lugović et al., 1990). The oldest loess in Bansko Brdo hill is detected at the Zmajevac locality where infrared optically stimulated luminescence (IRSL) indicated an age of 217 ± 22 ka, while the youngest loess has an IRSL age of 16.7 ± 1.8 ka (Galović et al., 2009).

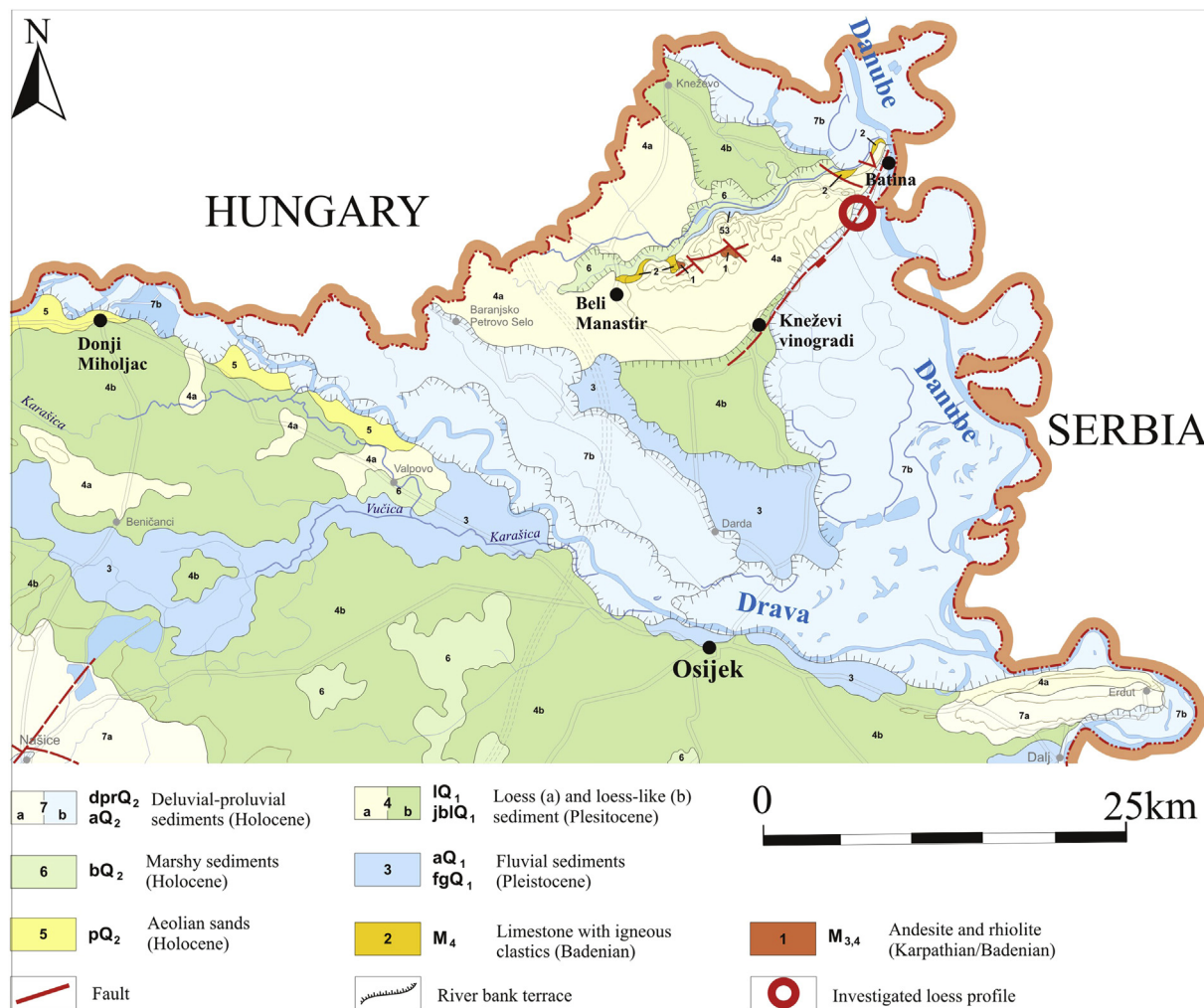


Fig. 2. Geological map (CGS, 2009) of Baranja region with encircled position of investigated Zmajevac LPS.

3. Methods

3.1. Field methods

Field investigation and sampling were carried out during winter months, because lush vegetation in spring and summer does not allow access to loess profiles. The aim was to include a maximum thickness of sediments. Bulk samples (8–10 kg) were collected from the Zmajevac (Zma) loess outcrop (Fig. 3) for all analysis. Field investigation included sedimentological logging, sampling, GPS positioning and photographing. Different horizons in loess were recognized and described. Sampling frequency was around 1.5 m.

3.2. Lab methods

Samples for grain – size analysis were decalcified with 5% HCL and dried in a heater for 24 h. Grain-size analyses combined wet sieving and the pipette method. Classification of the grain size distribution follows Wentworth (1922). Each loess sample was also sieved to obtain whole mollusc shells. Shells were derived from samples by screen-washing in distilled water, using a 0.7 mm mesh sieve. The identification of mollusc species follows taxonomic concepts of Ložek (1964) and Frank (2006). Assemblage analysis follows Ložek (1964).

Shells selected from each of the 13 samples for stable isotope analysis were cleaned in de-ionized water for the purpose of saving primary ratios of stable oxygen and carbon isotopes. Then they were crushed in powder. Obtained powder was heated to 500 °C for 30 min to remove the organic matter. Ratios were measured by automated continuous flow carbonate preparation GasBenchIII device and Thermo Electron Delta Plus XP mass spectrometer at the IAMC-CNR (Naples) isotope geochemistry laboratory. Acidification of samples was performed at 50 °C. For every six samples, an internal standard (Carrara Marble with $\delta^{18}\text{O} = 2.43\text{‰}$ versus VPDB and $\delta^{13}\text{C} = 2.43\text{‰}$ vs. VPDB) was run and for every 30 samples, the NBS19 international standard was measured. Standard deviations of carbon and oxygen isotope measures were estimated 0.1% and 0.08%, respectively, based on 20 measured samples, three times each. For statistical correctness and for obtaining results that will reflect paleoclimatic conditions in the best possible mode, for each analysis three mollusc shells from each of the two most common species were selected from 13 loess samples.

4. Zmajevac profile

Zmajevac LPS makes up a total of 18 m of loess (Fig. 3). It includes four palaeosols intercalated within five loess horizons and is defined as a typical loess. Detailed lithology of Zmajevac LPS, as well as its correlation with the chronological frame from MIS 6–MIS

ZMAJEVAC LPS (45° 48' 38" N, 18° 49' 7" E)

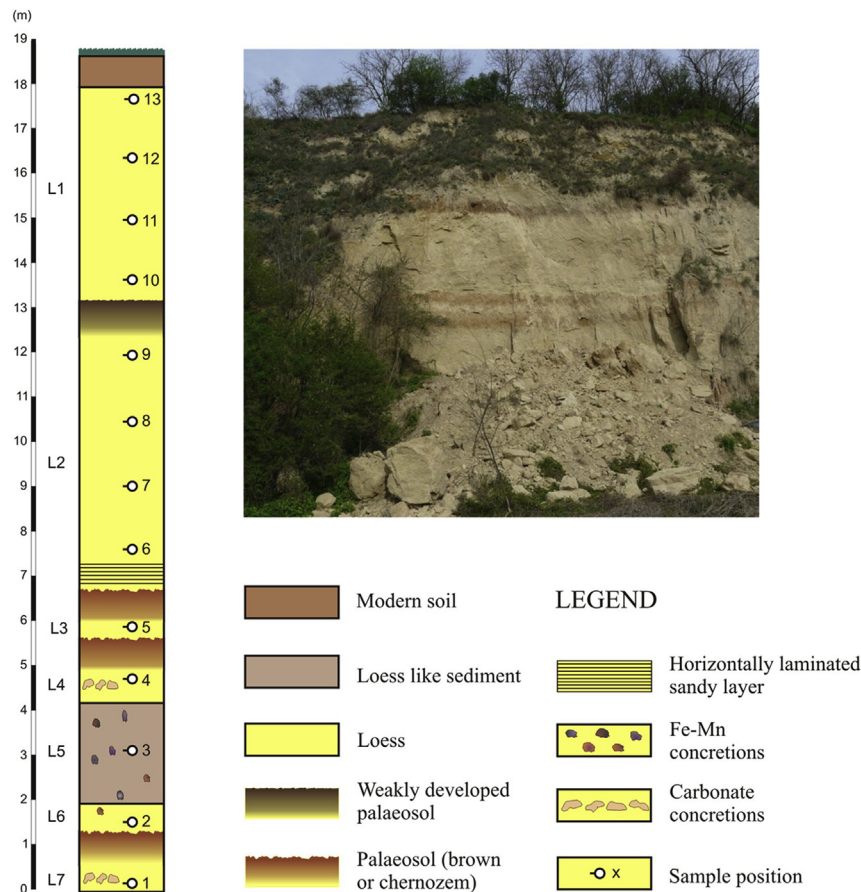


Fig. 3. Zmajevac LPS. Loess thickness is approx. 18 m. Three palaeosols are clearly visible at the top and in the middle of the profile photograph. The fourth, lowermost can be seen at the right, bottom corner of the photograph.

2, has been previously published in Galović et al. (2009) and Banak et al. (2012). Beyond that the latter authors investigated sediment petrography and mollusc palaeontology of the profile.

4.1. Grain-size distribution

Grain-size analysis in Banak et al. (2012) indicated silt as the dominant grain-size fraction in all 13 studied loess samples from the Zmajevac profile (Fig. 4). Average share of silt-size particles is 88.11%. Coarse-grained silt is the dominant silt fraction with average percentage of 41.38%. The laminated unit is composed of 81% sand, 11% silt and 8% clay, with a median grain-size of 0.22 mm. Skewness is fairly constant in all 13 samples, averaging of 0.79. Sorting is dominantly poor, with an average value 1.64.

The average CaCO_3 content in loess samples from Zmajevac LPS is 9.32%, with a maximum value of 23.3% that was recorded in sample No.10 and the minimum value of 2.9% in sample No.4. Modal analysis of the light (LMF) and heavy mineral fraction (HMF) was made on all 13 samples of loess. Light mineral fraction is dominated by quartz with an average share of 58.38%. Heavy mineral fraction is dominated by amphibole, garnet and epidote while chlorite is present in small percentages. Each of the three major groups of transparent heavy minerals is represented approximately in the range of 25–30%, and the relations between them vary. All of these aforementioned values point out that the Zmajevac profile is a typical loess profile from the Pannonian Basin.

4.2. Mollusc palaeontology

Mollusc palaeontology of the Zmajevac profile was documented by Banak et al. (2012), based on 13 loess samples with a total of 1705 terrestrial gastropod specimens bearing 13 species-level taxa (Fig. 5). Specimen richness, related to mollusc density within the samples varies in the loess profile. Therefore, the malacofaunal densities are moderate in horizons L7 (85) and L6 (117), strongly decreased in L5 (5) and L4 (7), and increased in L3 (90), L2 (213), and L1 (136). The identified mollusc species have been classified according to their palaeoenvironmental preferences following Kropp and Sümegei (1995) and Sümegei and Kropp (2002). The presence and abundance of each species in the sample is considered as a function of available palaeoclimate and palaeohumidity conditions and/or type of palaeovegetation. So, each sample is characterized by a certain malacofaunal assemblage. Terrestrial gastropod palaeoecology based on shells counted from the loess, documents cyclic transitions between cryophilous to cold resistant and mesophilous to thermophilous assemblage types. Species tolerating open and dry habitats are abundant in the Zmajevac profile. Five specific faunas which are all cold resistant were determined in the loess samples. *Helicopsis striata* fauna is the most dominant among five faunas.

The obtained quantified data allowed Banak et al. (2012) the definition of mollusc assemblages in the Zmajevac LPS for each loess horizon (from L1 to L7). The *Helicopsis striata* assemblage

ZMAJEVAC LOESS PROFILE (45° 48' 38" N, 18° 49' 7" E)

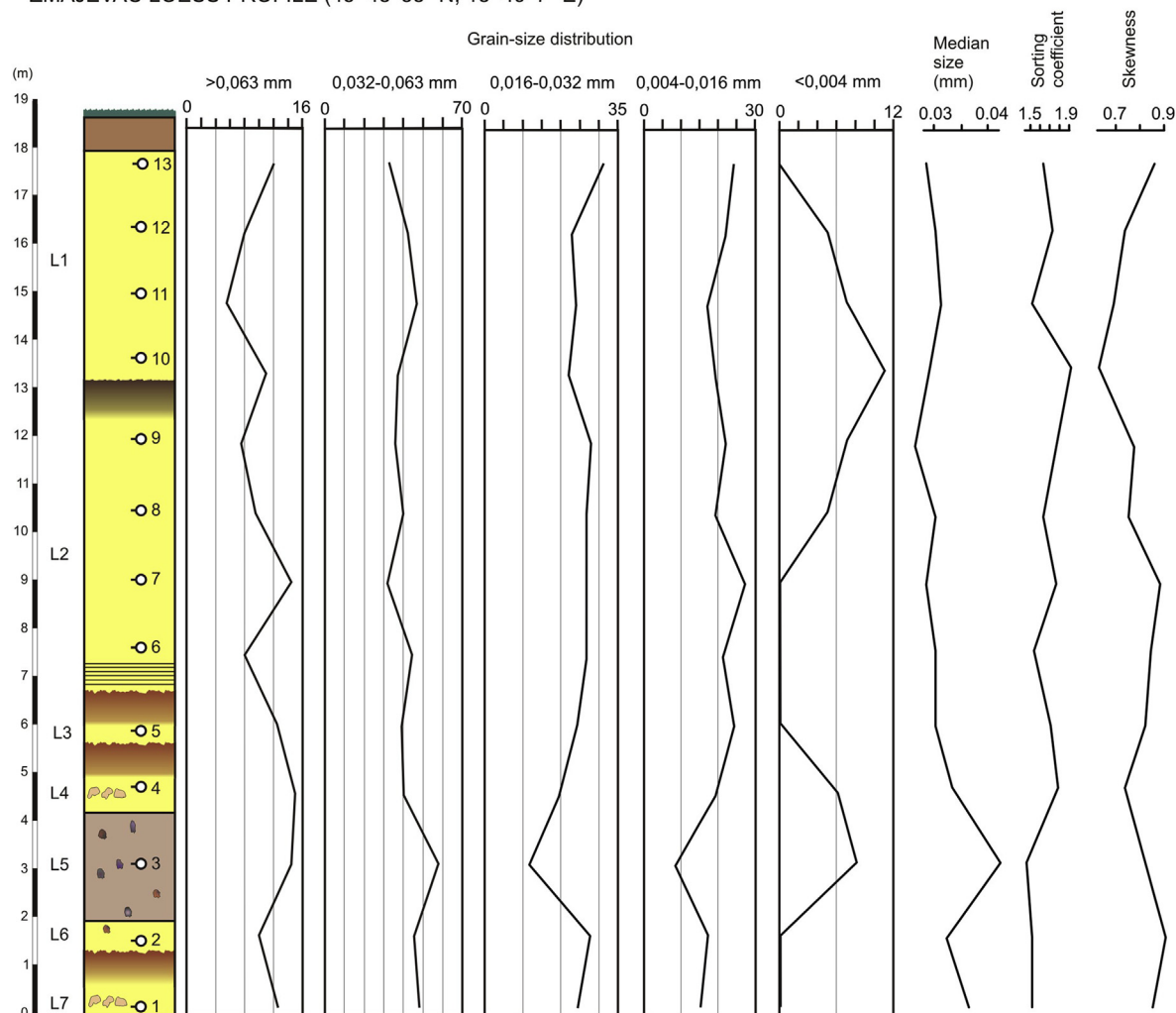


Fig. 4. Grain size distribution and coefficients in loess samples from Zmajevac LPS (modified after Banak et al., 2012).

indicates climate conditions preceding the last glacial/stadial maximum and it is characteristic assemblage of the “warm” loess environment in central Europe (Ložek, 1964). *Chondrula tridens* and *Arianta arbustorum* are generally abundant species, but their abundance is never high enough for definition of a pure *Chondrula tridens* or *Arianta arbustorum* assemblage. *Arianta arbustorum* species is cryogenic, hygrophilous species and assemblage is typical for humid forests, hillsides and lowland areas. *Chondrula tridens* species is a warm steppe species and assemblage is typical for interstadial mild, dry steppe – forest steppe. The *Pupilla loessica* and *Columella columella* assemblages are typical loess faunas and exponents of glacial/stadial maximum. Arid and cold climate conditions are indicated by the *Pupilla loessica* assemblage, while more humid conditions are represented by the *Columella columella* assemblage.

5. Stable isotope results

The two most common species which appear in all samples from the Zmajevac LPS were analysed. They are *Helicopsis striata* (WAGNER) and *Arianta arbustorum* (LINNAEUS). Stable oxygen isotope values for these two species range from –5.76‰ to –2.45‰ (Fig. 6). Stable carbon isotope values range

from –8.83‰ to –6.84‰ (Fig. 6). Average value for $\delta^{18}\text{O}$ is –4.20‰ and average value for $\delta^{13}\text{C}$ is –7.74‰. It is clear that $\delta^{13}\text{C}$ values varies to a lesser extent than $\delta^{18}\text{O}$ values. The largest number of measured $\delta^{18}\text{O}$ values is characterized with malacofauna resistant to cold glacial and/or stadial conditions and $\delta^{18}\text{O}$ values range from –2.8‰ to –4.4‰. In that group $\delta^{13}\text{C}$ values are in range from –6.84‰ to –8.83‰. A certain number of measured $\delta^{18}\text{O}$ values which are more negative than –4.4‰ can be attributed to cryophilic mollusc assemblage types which prefer stadial conditions. $\delta^{18}\text{O}$ values more positive than –2.5‰ could correspond to mesophilic and thermophilic mollusc assemblage types. The values in each sample vary additionally, depending on which mollusc species is analysed. Regularity, which would indicate that one analysed mollusc species shows constantly lower or higher $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values in relation to the other analysed species, is not registered. Stable oxygen isotope values from four uppermost samples range from –2.45‰ to –4.41‰ (Table 1). Average value is –3.69‰. Stable carbon isotope values from four uppermost samples range from –6.84‰ to –7.75‰ (Table 1). Average value is –7.25‰. The values for stable oxygen isotope are in accordance with results from other loess profiles in the Pannonian Basin with emphasis on MIS 2 stage (Kehrwald et al., 2010).

Table 1
Stable isotope values measured in mollusc shells at Zmajevac LPS.

Sample	$\delta^{18}\text{O}$	$\delta^{13}\text{C}$
13	-2.45	-7.45
12	-3.84	-6.96
11	-4.05	-6.84
10	-4.41	-7.75
9	-3.26	-8.83
8	-3.16	-7.47
7	-4.45	-7.31
6	-5.15	-8.82
5	-5.27	-8.31
4	-3.57	-8.20
3	-4.13	-7.49
2	-5.18	-7.91
1	-5.76	-7.23

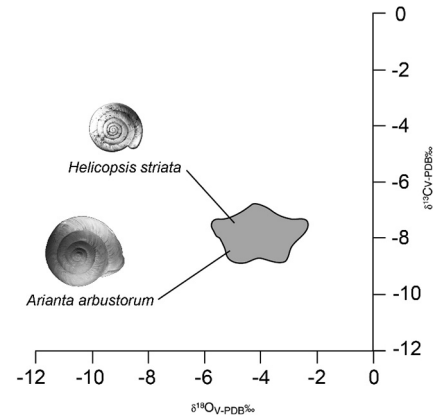


Fig. 6. Stable isotope values for two most abundant mollusc species sampled in Zmajevac LPS.

6. Discussion

6.1. Comparison of mollusc assemblages with others in Pannonian Basin

Previous malacofaunal research conducted on loess profiles in Eastern Croatia (Poje, 1982, 1986) show significant concordance of results with the results obtained by Banak et al. (2012) from the Zmajevac LPS, which have been discussed in detail. The mollusc species distribution of the Zmajevac LPS shows certain similarities also with the Irig loess profile on the southern slope of Fruška Gora Mt. in NW Serbia (Marković et al., 2007). *Chondrula tridens* and *Helicopsis striata* assemblages which dominate in the Irig loess profile are also present in the Zmajevac LPS but *Chondrula tridens* is not so dominant in Zmajevac LPS. Also, in contrast to the lowermost part of the Irig loess profile where *Vallonia costata* and *Clausilia dubia* species are present, two lowermost horizons in the Zmajevac LPS bear cooler climate representatives in form of *Columella columella* fauna, reflecting the Middle Pleistocene Penultimate Glacial (MIS 6) conditions. Based on this differences we can conclude that these two loess profiles basically differ from each other.

On the Erdut loess profile (Poje, 1982) determined malacofauna was detected in four loess horizon. The base horizon is characterized by *Columella columella* assemblage, while in the three remaining horizons *Helicopsis striata* assemblage, with elements of other faunas, is dominant. These results are comparable to the results obtained from the Zmajevac LPS on Bansko Brdo hill. Still, there are some notable, minor differences in faunal content. The L4, L5 and L6 horizons of the studied Zmajevac profile with dominant *Helicopsis striata* assemblage show strong influence of *Arianta arbustorum* and *Columella columella* assemblage, whereas in the profile that Poje (1982) studied, it displayed a dominance of *Helicopsis striata* assemblage with influence of *Chondrula tridens* assemblage. The Late Pleistocene assemblages from the Petrovaradin loess profile in NW Serbia show colder and more humid conditions than in either the Irig or Zmajevac LPS, probably due to the palaeogeographic position at the northern slope of Fruška Gora Mt. (Marković et al., 2005, 2008). The topmost L1 horizon from this study and the topmost horizon from the Erdut profile (Poje, 1982) display great similarity, because the dominant *Helicopsis striata* assemblage is accompanied with a *Chondrula tridens* assemblage. The faunal assemblages from that interval at Zmajevac LPS display a dominance of transitional palaeoecological assemblage types, whereas uniformly defined types are rare (Banak et al., 2012). The lowermost horizons at Zmajevac LPS (L7 and L6) and from the Erdut profile (Poje, 1982) display difference which is marked by dominant

ZMAJEVAC LOESS PROFILE (45° 48' 38" N, 18° 49' 7" E)

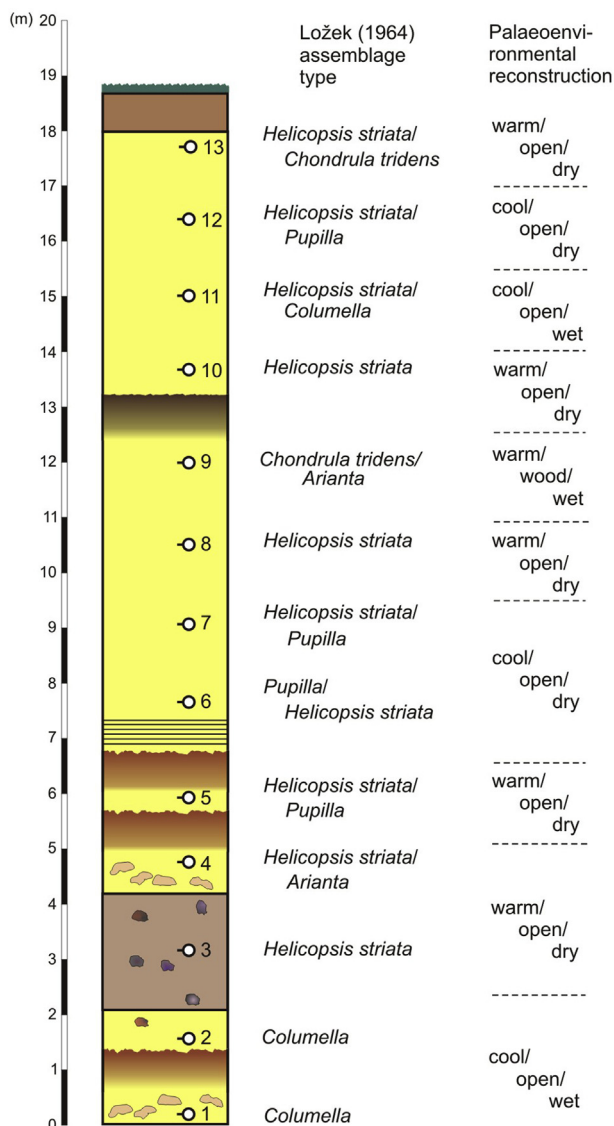


Fig. 5. Mollusc assemblages determined in loess samples from Zmajevac LPS (modified after Banak et al., 2012).

Helicopsis striata assemblage at the Erdut profile and *Columella columella* assemblage at Zmajevac LPS. The recorded mollusc assemblages display small but important differences to other Pannonian basin LPS, especially for horizons L7, L6 and L3. All this compared results of malacofaunal assemblages from nearby loess profiles suggests that climate conditions that dominated this region were similar, with some differences which were a result of palaeogeographically and palaeogeomorphically driven microclimate conditions. The fauna from middle and upper horizons (L3, L2 and L1) of the Zmajevac LPS displays certain similarity with Madaras loess section in South Hungary (Hupuczi and Sümege, 2010), although differences prevail. L3 horizon in the Zmajevac LPS and K L1 LL3 horizon from Madaras both contain *Helicopsis striata* assemblage. L2 horizon from the Zmajevac LPS differs from K L1 LL2 horizon at Madaras section, because *Vallonia costata* and *Pupilla muscorum* species dominate in this loess profile while in the Zmajevac LPS *Pupilla muscorum* is present but not dominant and *Vallonia costata* is not present at all. L1 horizon in the Zmajevac LPS differs from K L1 LL1 because *Helicopsis striata* and *Chondrula tridens* assemblages dominate here and *Columella columella* species is scarce, while *Columella columella* and *Vallonia tenuilabris* species dominate in Madaras.

6.2. Late Pleistocene climate reconstruction based on $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values

Stable isotope ratios of $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ were measured from fossil *Helicopsis striata* and *Arianta arbustorum* shells. Modern European land snails are active in the 10–27 °C temperature range and hibernate or become inactive at temperatures below 10 °C (Heller, 1990; Thompson and Cheny, 1996). This implies that stable isotope ratios recorded in mollusc shells represent a warmer period in continental climate when snails formed their shells. This period from spring to fall can be 160–210 days long (Frömming, 1954; Ant, 1963) and it displays an average growing season temperature – AGS temperature. It is the same with fossil snails. Also, snails are active in building their shells during and after the rain (Ward and Slowtow, 1992). This is important because it provides a direct link from rainwater $\delta^{18}\text{O}$ values and $\delta^{18}\text{O}$ values measured in the mollusc shells. The relationship between the $\delta^{18}\text{O}$ of meteoric water and the values of land-snail shells has been studied for over 30 years (Yapp, 1979; Lecolle, 1985; Goodfriend and Magaritz, 1987; Goodfriend, 1992, 1999). Variations in land snail shell $\delta^{18}\text{O}$ is a function of temperature, relative humidity, $\delta^{18}\text{O}$ of water vapor, and $\delta^{18}\text{O}$ of liquid water ingested by the snail (Lecolle, 1985; Goodfriend et al., 1989; Balakrishnan and Yapp, 2004). Intra-shell variation of values measured in LGM snails ranges from 0‰ to –5.5‰ (Kehrwald et al., 2010). Therefore whole shells were crushed and analysed in this study to avoid this strong variation. The $\delta^{18}\text{O}$ value in mollusc shells is enriched on average by 5‰ relative to equilibrium with rainwater (Lecolle, 1985; Goodfriend and Magaritz, 1987). This means that a $\delta^{18}\text{O}$ value of palaeo – rainwater incorporated in a mollusc shell which displays a $\delta^{18}\text{O}$ value of approx. –2‰ was approx. –7‰. To compare climate conditions in the Pleistocene with recent ones it is necessary to know the $\delta^{18}\text{O}$ values of recent meteoric water and recent AGS temperatures from the same or nearby region. The nearest measured $\delta^{18}\text{O}$ value of rainwater to Zmajevac LPS is recorded in Zagreb, Croatia, –6.11‰ for summer months – June, July, August (JJA) in the last two decades (Kehrwald et al., 2010). This values represent shorter period than AGS, but it is the closest approximation. In the last two decades mean JJA temperature recorded in Zagreb was +19.7 °C. This temperature is almost identical to mean July temperatures from MIS 3 stage palaeosols from Hungary and Serbia LPS's determined by using the malacothermometer method. This method yielded

mean July paleotemperatures of 19–21 °C (Krolopp and Sümege, 1995; Marković et al., 2007; Novothny et al., 2011).

If we compare $\delta^{18}\text{O}$ values from Zmajevac LPS mollusc shells, enlarged for 5‰, we can clearly see that $\delta^{18}\text{O}$ values of meteoric water in the Late Pleistocene ranged from approx. –7.45‰ to approx. –10.76‰. Compared with the $\delta^{18}\text{O}$ value of –6.11‰ from present JJA measurements, this suggests that $\delta^{18}\text{O}$ values were constantly lower/more negative and consequently that AGS temperatures during the Late Pleistocene were much lower than present. The mean annual $\delta^{18}\text{O}$ value for Zagreb is –8.33‰ (Kehrwald et al., 2010) and mean annual temperature (MAT) for Zagreb in last two decades is +12 °C. As most of the samples from Zmajevac LPS display more negative $\delta^{18}\text{O}$ values than –8.33‰, it is reasonable to conclude that even the AGS temperatures during the Late Pleistocene were lower than the present MAT in Zagreb.

Based on IRSL age estimate of 20.2 ± 2.1 ka, Galović et al. (2009) defined the uppermost part of the LPS at Zmajevac as Upper Pleniglacial MIS 2 stage. It corresponds with the L1 loess horizon of this study. Four loess samples were retrieved from this horizon and malacofaunal assemblages from these samples represent the climate and environment from MIS2. Therefore, these results can be correlated with $\delta^{18}\text{O}$ values from snail shells from the Pannonian Basin (Kehrwald et al., 2010), Southern Italy (Colonese et al., 2007) and North America (Balakrishnan et al., 2005) which are also a part of MIS2 (Fig. 7).

Kovács et al. (2012) estimated the MAT for MIS 2 before the LGM peak, reconstructed from oxygen isotopes in mammoth tooth enamel, in a range from 6.2 °C to 11.2 °C in the Czech Republic and Slovakia. This is north of Zmajevac LPS but it can serve as a marker, if we assume the possible decrease in temperature due to latitude increase towards the north. Schatz et al. (2014) estimated temperatures of 6.7 °C (MS based), 8.5 ± 0.6 °C (XRF-1 based) and 8.9 ± 4.4 °C (XRF-2 based) in Northern Hungary for MIS 2 stage. Corcho Alvarado et al. (2011) calculated a MAT of 4.5 °C using noble gas thermometry (NGT) in Central Europe which is significantly lower than the aforementioned results.

Changes of $\delta^{18}\text{O}$ values in fossil shells reflect paleotemperature changes over time. This is useful if we want to reconstruct paleotemperature changes over longer period of time, and not just for

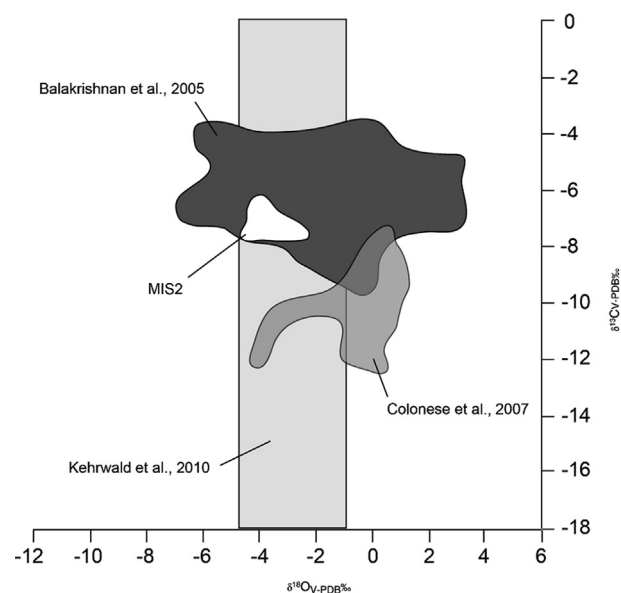


Fig. 7. Comparison of results for MIS2 from Zmajevac LPS with ones obtained from other European and North American Late Pleistocene mollusc assemblages.

ZMAJEVAC LOESS PROFILE (45° 48' 38" N, 18° 49' 7" E)

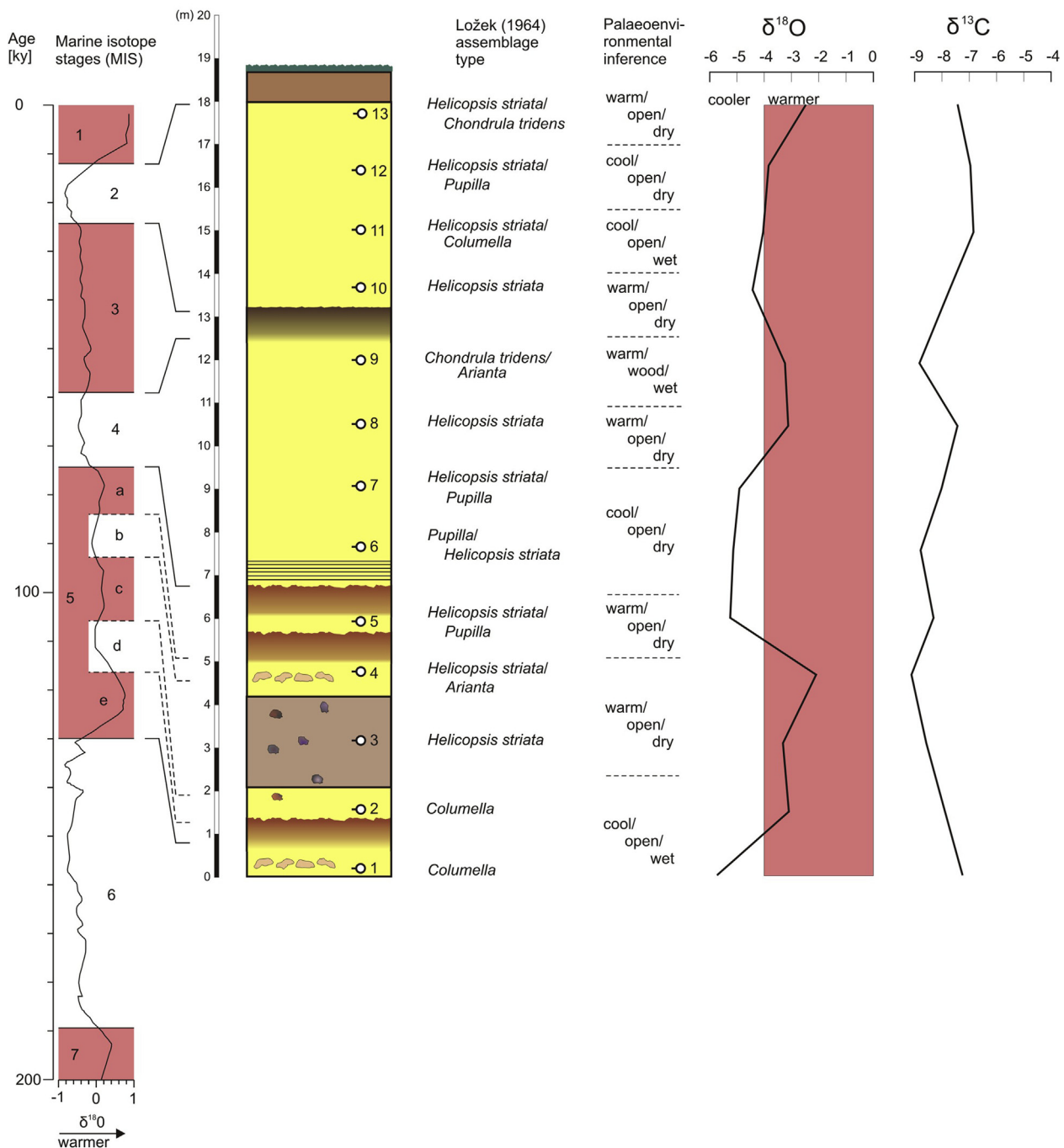


Fig. 8. Malacofaunal assemblage types from Banak et al. (2012) correlated with stable isotope results of the present study. Time frame and MIS stages are given by Galović et al. (2009) and Banak et al. (2012).

MIS 2. If the $\delta^{18}\text{O}$ value changes by 0.5‰ it reflects a paleotemperature change of $\approx 2^\circ\text{C}$ (Goodfriend, 1999). The formula for calculating AGS paleotemperature changes in the Zmajevac LPS, adjusted according to Goodfriend (1999), would be:

$$\Delta T(^{\circ}\text{C}) = \left(\delta^{18}\text{O}_{\text{max.}} - \delta^{18}\text{O}_{\text{min.}} / 0.5\text{‰} \right) \times 2^{\circ}\text{C}$$

where:

ΔT – is: relative paleotemperature change

$\delta^{18}\text{O}_{\text{max}}$ – is: maximal $\delta^{18}\text{O}$ shell value for Zmajevac LPS (–2.45)

$\delta^{18}\text{O}_{\text{min}}$ – is: minimal $\delta^{18}\text{O}$ shell value for Zmajevac LPS (–5.76)

If we enter our $\delta^{18}\text{O}$ values in this formula AGS paleotemperature changes through the Late Pleistocene in the Zmajevac LPS would range by 13.24°C .

Others set different ratios and interdependence of $\delta^{18}\text{O}$ values and temperature. If the $\delta^{18}\text{O}$ value changes by 0.35‰ it reflects a paleotemperature change of $\approx 1^\circ\text{C}$ (Balakrishnan et al., 2005). The

formula adjusted according to their research would be slightly altered:

$$\delta^{18}\text{O}(\text{‰}) = (\delta^{18}\text{O}_{\text{max.}} - \delta^{18}\text{O}_{\text{min.}}) / 0,35\text{‰} \times 1^\circ\text{C}$$

If we enter the same $\delta^{18}\text{O}$ values in this formula AGS paleo-temperature changes through the Late Pleistocene in the Zmajevac LPS would range by 9.457°C . If we compare this results with MAT temperatures for other Pannonian Basin LPS, it is plausible to conclude that the second formula and the range of 9.457°C are more accurate.

Both of these values suggest strong and constant changes during the Late Pleistocene in the Baranja region which are recorded in mollusc shells from the Zmajevac LPS. The exact temperature range can not be determined because the flux-balance model, or physical and chemical processes during which isotope fractionation occurs in mollusc shells is complicated and subject to the influence of many factors.

The isotope curve for $\delta^{18}\text{O}$ values from Zmajevac LPS displays some deviation in regards to palaeotemperatures or palaeoclimatic conditions assigned to warm/cold relation and represented by malacofaunal assemblages (Fig. 8). These deviations in particular are probably a result of the same complex flux–balance model between water used by the mollusc and its shell, which does not respond with the same intensity and to palaeotemperature changes (Balakrishnan and Yapp, 2004).

Climate changes during the glacial and interglacial periods are the main cause for changes in vegetation which are reflected in $\delta^{13}\text{C}$ values of plants (Huang et al., 2001; Hall and Penner, 2013). Therefore, stable carbon isotopes or $\delta^{13}\text{C}$ values can be used to determine the diet of land snails, which can help in palaeoenvironmental reconstruction. The $\delta^{13}\text{C}$ value of atmospheric CO_2 does not affect the $\delta^{13}\text{C}$ value of snail shells, so these values are a relevant indicator of snail diet (Balakrishnan and Yapp, 2004). If the $\delta^{13}\text{C}$ values are more negative, it is an indication that the molluscs consumed C3 plants in their diet and that the climate was cooler and more humid (Goodfriend and Ellis, 2002; Yanes et al., 2008). If the $\delta^{13}\text{C}$ values are more positive, it is an indication that the snails consumed C4 plants in their diet, which indicates a more arid environment (Galy et al., 2008; Yanes et al., 2008).

A study of Sümeği et al. (2012) shows that relatively stable woodland – grassland ecotone with fluctuating proportion of arboreal elements and taxa was the dominant vegetation type in the Pannonian Basin between 140 and 16 kys, a time span that covers this study as well. This woodland – grassland ecotone was preserved even during the strongest cooling, during phases of a treeless steppe (Sümeği et al., 2012). In this mixture of temperate, arctic and alpine ecosystems C3 plants typically dominate (Schatz et al., 2011). Soils formed in Hungary Tokaj region – Pannonian Basin, display $\delta^{13}\text{C}$ values in narrow range from -24‰ to -25‰ (Schatz et al., 2011) which is typical for soils developed under plants using the C3 photosynthetic pathway (Ambrose and Sikes, 1991). The variation of $\delta^{13}\text{C}$ values measured in shells from loess samples of the Zmajevac LPS is also modest and ranges from -8.83‰ to -6.84‰ . Therefore C4 vegetation as a diet source for the Zmajevac LPS molluscs can be excluded because these plants display very different $\delta^{13}\text{C}$ values, ranging from -8‰ to -16‰ (Ambrose and Sikes, 1991) which is contrary to our results. $\delta^{13}\text{C}$ values measured in the fossil mollusc shells are enriched by $8\text{--}19\text{‰}$ compared to the values of the plants that they ingested (McConnaughey and Gilić, 2008). This means that $\delta^{13}\text{C}$ values of plants that mollusc from Zmajevac LPS ingested, were approximately from -14.84‰ to -16.83‰ if we use minimal 8‰ enrichment. That is very close to the most negative margin for C4 plants.

In case of maximal 19‰ enrichment these values are in range from -25.84‰ to -27.83‰ .

Research of Marković et al. (2007), Novothny et al. (2011), Schatz et al. (2011) and Sümeği et al. (2012) compared with results from this study suggests that for the entire time span during which the Zmajevac LPS was accumulated, C3 plants have been the main vegetation type. This indicates that late Pleistocene climate in the Baranja region was similar to other regions in the Pannonian Basin with some differences which can be attributed to local geomorphology and microclimate conditions.

7. Conclusion

Stable oxygen values show significant palaeotemperature changes during the Late Pleistocene in the Baranja region with average growing season (AGS) temperature change in the range of 13.24°C (Goodfriend, 1999) or 9.457°C (Balakrishnan et al., 2005). The second calculated temperature range is more plausible. The overall climate was much cooler than present. Stable carbon isotope values show that for the entire time span during which the Zmajevac LPS was accumulated, C3 plants most probably have been the main vegetation type. These include trees and cold steppe grasses. Comparison of malacofaunal assemblages from the Zmajevac LPS to other LPS from the Pannonian Basin suggests that climate conditions that dominated this region were similar, with some differences that were a result of geomorphology and specific microclimate conditions.

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